Bert Baumgaertner

CONTACT INFORMATION

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EMPLOYMENT

University of Idaho, Moscow, Idaho

Professor of Philosophy	2025 – present
Modeling Core Co-Director, **IMCI**	July 2024 – present
Associate Professor of Philosophy	2019 – 2025
Department Chair	January 2023 – June 2024
Acting Chair	August 2020 – December 2020
Assistant Professor of Philosophy	2013 – 2019

EDUCATION

University of California, Davis, California, USA

Ph.D, Philosophy (2013)

Dissertation "Vagueness Judgments: An Information-Theoretic Approach"

Wilfrid Laurier University, Waterloo, Ontario, Canada

B.A, Philosophy (2007)

RESEARCH INTERESTS

Areas of Specialization

Cognitive Science, Computational Philosophy, Social Epistemology

Areas of Competence

Logic, Game Theory, Decision Theory, Applied Ethics (AI and Data Ethics)

PUBLICATIONS (REFEREED)

Note: Two-authored papers are equal contributions. Papers with three or more, **bold** indicates lead/senior author. Undergraduate authors are indicated with *.

35. Chakrabarti, L., Johnson-Leung, J., Baumgaertner, B., Vakanski, A., Xian, M., Zhang, B. (2026) "Post-Hoc Understanding of Metaphor Processing in Decoder-Only Language Models via Conditional Scale Entropy" In *Pattern Recognition. ICPR 2026 International Workshops*, Lecture Notes in Computer Science, Springer, Cham, 524–539. https://doi.org/10.1007/978-3-032-39388-3_37

34. Snow, Jacqueline; Zajchowski, Chris A.B.; Karl, Jason; Baumgaertner, Bert (2026) "Signs of Intelligence: Trust in Generative Artificial Intelligence for Conservation Messaging" *Parks Stewardship Forum: The Interdisciplinary Journal of Place-Based Conservation*, 42(3). https://doi.org/10.5070/p5.69182

33. Tovissode, C.F., Justwan, F., Baumgaertner, B. (2026) "Patterns between evidence seeking behaviors, reasoning, and cognitive reflection: A supervised clustering approach" *PLoS One* 21(6): e0352096. https://doi.org/10.1371/journal.pone.0352096

32. Justwan, Florian; Baumgaertner, Bert; McBrayer, Markie; Bindley, Jacob*; Benitez, Alexa*; Haile, Kira*; Kludt-Painter, Bergen* (2026) "Evaluating Evidentiary Standards in the Realm of Citizen Policy Evaluation" *Social Science Quarterly*, https://doi.org/10.1111/ssqu.70128

31. Justwan, F., Berejikian, J.D., Baumgaertner, B., (2026) "The Causes and Consequences of Public Misperceptions about the Origins of the Russo-Ukrainian War" *International Political Science Review* (online Jan 25, 2026) https://doi.org/10.1177/01925121251403578

30. Justwan, Florian; Baumgaertner, Bert (2025) "The effects of ideology and cognitive reflection on evidence gathering behavior in the political domain" *PLoS One* 20(12): e0338088. https://doi.org/10.1371/journal.pone.0338088

29. Tovissode, C., Baumgaertner, B. (2024) "Heterogeneous risk tolerance, in-groups and epidemic waves" *Front. Appl. Math. Stat.*, Sec. *Mathematical Biology*, vol 10 - 2024. https://doi.org/10.3389/fams.2024.1360001

28. Bert Baumgaertner, Zeth duBois (2024) "Search is a Hammer, Generative Chat is a Loom; Beware the Technological Attribution Error" In: Coman, A., Vasilache, S. (eds) *Social Computing and Social Media.* HCII 2024. Lecture Notes in Computer Science, vol 14703. Springer, Cham. https://doi.org/10.1007/978-3-031-61281-7_22

27. Baumgaertner, B., Lassiter, C. (2024) "Precedent and rest stop convergence in reflective equilibrium" *Synthese* 203, 88. https://doi.org/10.1007/s11229-024-04499-x

26. Baumgaertner, B., & Lassiter, C. (2023) "Convergence and Shared Reflective Equilibrium" *Ergo* 10:24 https://doi.org/10.3998/ergo.4654

25. Buzbas EO, Devezer B, Baumgaertner B. (2023) The logical structure of experiments lays the foundation for a theory of reproducibility. *R. Soc. Open Sci.* 10: 221042. https://doi.org/10.1098/rsos.221042

24. Baumgaertner, B., Justwan, F. (2022) "The preference for belief, issue polarization, and echo chambers." *Synthese* 200, 412 (2022). https://doi.org/10.1007/s11229-022-03880-y

23. Justwan, Florian; Baumgaertner, Bert; Curtright, Madeleine* (2022) "Meddling in the 2016 Elections and Satisfaction with Democracy in the US" *Political Studies* https://doi.org/10.1177/00323217221126300

22. Rebecca C. Tyson, Noah M. Marshall*, Bert Baumgaertner (2022) "Transient prophylaxis and multiple epidemic waves" *AIMS Mathematics*, 7(4):5616-5633. https://doi.org/10.3934/math.2022311

21. McBrayer, M., Baumgaertner, B., Justwan, F. (2022) "The Effects of COVID-19 on Political Efficacy" *American Politics Research*, 50(1): 97-107. https://doi.org/10.1177/1532673X211041322

20. **Baumgaertner, B.,** Benjamin J. Ridenhour, Florian Justwan, Juliet E. Carlisle, Craig R. Miller (2020) "Risk of disease and willingness to vaccinate in the United States: a population-based survey" *PLOS Medicine*, 17(10): e1003354 https://doi.org/10.1371/journal.pmed.1003354

19. Majmudar, J.R., Krone, S.M., Baumgaertner, B.O., Tyson, R.C. (2020) "Voter models and external influence" *The Journal of Mathematical Sociology*, Vol.44, No.1, (Online June 14, 2019), https://doi.org/10.1080/0022250X.2019.1625349

18. Tyson, R., Baumgaertner, B., Hamilton, S.*, Lo, A., Krone, S. (2020) "The timing and nature of behavioural responses affect the course of an epidemic" *Mathematical Bulletin of Biology*, Vol.82, No.14 (Online January 14, 2019) https://doi.org/10.1007/s11538-019-00684-z

17. Florian Justwan, Bert Baumgaertner, Juliet E. Carlisle, Emma Carson*, Jordan Kizer* (2019) "The effect of trust and proximity on vaccine propensity" *PLoS ONE* 14(8): e0220658. https://doi.org/10.1371/journal.pone.0220658

16. Devezer, B., Nardin, LG., Baumgaertner, B., Buzbas, E. (2019) "Scientific discovery in a model-centric framework: Reproducibility, innovation, and epistemic diversity" *PLoS ONE*, Vol. 14, No. 5, e0216125, (Online May 15, 2019) https://doi.org/10.1371/journal.pone.0216125

15. **Baumgaertner, B.,** Fetros, P.*, Tyson, R., Krone, S. (2018) "Spatial Opinion Dynamics and the Effects of Two Types of Mixing" *Physical Review E*, Vol. 98, No. 2, 022310-23, (Online August 13, 2018) DOI 10.1103/PhysRevE.98.022310.

14. **Baumgaertner, B.**, Carlisle, J., Justwan, F. (2018) "The Influence of Political Ideology and Trust on Willingness to Vaccinate" *PLoS ONE*, Vol. 13, No. 1, e0191728, https://doi.org/10.1371/journal.pone.0191728

13. Justwan, F., Baumgaertner, B., Carlisle, J., Clark, A., Clark, M. (2018) "Social Media Echo Chambers and Satisfaction with Democracy among Democrats and Republicans in the Aftermath of the 2016 U.S. Elections" *Journal of Elections, Public Opinion & Parties*, (Online Feb 8, 2018) DOI 10.1080/17457289.2018.1434784.

12. **Baumgaertner, B.** (2018) "Models of Opinion Dynamics and Mill-style Arguments for Opinion Diversity" *Historical Social Research (Special Issue: Agent-Based Modeling in Social Science, History, and Philosophy)*, 43 (1): 210-33. https://doi.org/10.12759/hsr.43.2018.1.210-233

11. **Baumgaertner, B.** and Holthuijzen, W.* (2017) "Non-epistemic Values and Concerns about Evolutionary Mindsets in Conservation Policy: A Response to Kareiva and Fuller's 'Beyond Resilience' " *Global Policy*, (Online July 17, 2017) DOI 10.1111/1758-5899.12469.

10. **Baumgaertner, B.** and Holthuijzen, W.* (2016) "On Non-Epistemic Values in Conservation Biology" *Conservation Biology*, Vol. 31, No. 1, 48-55.

9. Roe, S. and **Baumgaertner, B.** (2016) "Extended Mechanistic Explanations: Expanding the current mechanistic conception to include more complex biological systems" *Journal for General Philosophy of Science*, Vol. 48, No. 4, 517-534. DOI 10.1007/s10838-016-9356-6.

8. Luis G. Nardin, Craig R. Miller, Benjamin J. Ridenhour, Stephen M. Krone, Paul Joyce, **Bert O. Baumgaertner** (2016) "Planning horizon affects prophylactic decision-making and epidemic dynamics" *PeerJ*, 4:e2678, https://doi.org/10.7717/peerj.2678

7. **Baumgaertner, B.**, Tyson, R., Krone, S. (2016) "Opinion strength influences the spatial dynamics of opinion formation" *The Journal of Mathematical Sociology*, Vol. 40, No. 4, 207-218. DOI 10.1080/0022250X.2016.1205049.

6. Banzhaf, W., Baumgaertner, B., Beslon, G., Doursat, R., Foster, J.A., McMullin, B., Veloso de Melo, V., Miconi, T., Spector, L., Stepney, S., White, R., (2016) "Defining and Simulating Open-Ended Novelty: Requirements, Guidelines, and Challenges" *Theory in Biosciences*, Vol. 131, No. 3, 131-161. DOI 10.1007/s12064-016-0229-7.

5. **Baumgaertner, B.** (2014) "Yes, No, Maybe So: A Veritistic Approach to Echo Chambers Using a Trichotomous Belief Model" *Synthese*, Vol. 191, No. 11, 2549-2569.

4. **Baumgaertner, B.** and Weiss, A. (2014) "Do Emotions Matter in the Ethics of Human -Robot Interaction? – Artificial Empathy and Companion Robots" In *Proceedings of the 3rd International Symposium on New Frontiers in Human-Robot Interaction.*

3. **Baumgaertner, B.** (2013) "Smooth yet Discrete: Modeling both Non-Transitivity and the Smoothness of Graded Categories with Discrete Classification Rules" *Minds and Machines*, Vol. 24, No. 3, 353-370.

2. **Baumgaertner, B.**, Fernández, R., Stone, M. (2012) "Towards a Flexible Semantics: Colour Terms in Collaborative Reference Tasks" In *Proceedings of the First Joint Conference on Lexical and Computational Semantics (*SEM)*, Montreal, Canada.

1. **Baumgaertner, B.** (2012) "Vagueness Intuitions and the Mobility of Cognitive Sortals" *Minds and Machines*, Vol. 22, No. 3, 213-234.

Chapters

1. **Baumgaertner, B.** (Lead Guest Editor) and Luciano, F. (2016) "Introduction: The Philosophy of Information" in Special Issue of *Topoi: An International Review of Philosophy* (Online) No. 35, 157-159.
2. **Baumgaertner, B.** (2013) "Mind" Chapter in *Philosophy of Information: A Simple Introduction* (available online: https://socphilinfo.github.io/PI-Intro/)

Invited & Reviews

1. Baumgaertner, Bert (2023) "Tanja Rechnitzer: Applying Reflective Equilibrium" (Book Review) *Ethical Theory and Moral Practice*, DOI 10.1007/s10677-023-10365-5
2. Baumgaertner, Bert (2019) "Groupstrapping, Bootstrapping, and Oops-strapping: A Reply to Boyd." *Social Epistemology Review and Reply Collective* 8 (6): 4-7.https://wp.me/p1Bfg0-4bE

Software & Data

1. Baumgaertner, B. and Potter, N. "Good Thinking Corpus" Training corpus (27,252 records, 182-code taxonomy) for critical thinking, logic, decision theory, and game theory education. HuggingFace: https://huggingface.co/datasets/bertybaums/good-thinking-corpus. DOI: 10.5281/zenodo.19324917
2. Baumgaertner, B. "NARC: Narrative Augmented Reasoning Challenges" 633 hand-authored puzzles (226 with human-written narratives) and an ablation battery separating lexical, order and form dependence; evaluated on eight open-weight models and one closed model. Paper under review at ICLR 2027. Website: https://narc.insight.uidaho.edu
3. Baumgaertner, B. "Narrative Hierarchy Model (NHM)" Dual-surface grammar in which every event rewrites into a sentence and a grid transition; 21,864 NARC puzzles generated from TinyStories with the narrative-necessity property certified by construction, plus a simulator (grammar × convention × symmetry × mask) and an attribution-error grammar. Data and generator release with the ICLR paper.
4. Baumgaertner, B. "MARC2: Metaphor Abstraction and Reasoning Corpus v2" Figurative language puzzles from ARC-AGI2 (16,711 rows). HuggingFace: https://huggingface.co/datasets/bertybaums/marc2. DOI: 10.5281/zenodo.19241782
5. Baumgaertner, B. "MARC: Metaphor Abstraction and Reasoning Corpus" 78 puzzles with 1,230 figurative descriptions and LLM behavioral trial data. HuggingFace: https://huggingface.co/datasets/bertybaums/marc
6. Baumgaertner, B. "MARC-from-LARC" Automated generation and verification pipeline for metaphor-based reasoning puzzles. GitHub: https://github.com/bertybaums/marc-from-larc
7. Baumgaertner, B. "CX-Bot" Corpus of 4,474 philosophical counterexamples across 16 domains for training language models. GitHub: https://github.com/bertybaums/cx-bot
8. Baumgaertner, B. "Code Names" AI vs. AI Codenames tournament engine — LLMs and embedding agents play Codenames, rated by Elo. GitHub: https://github.com/bertybaums/code-names

Pedagogical Materials

1. Baumgaertner, B. and Research Computing and Data Services, "Cluer: A CodeNames-style Game for AI Literacy about Semantic Embeddings" Supported by Institute for Interdisciplinary Data Sciences, and by College of Letters, Arts, and Social Sciences, U. of Idaho. https://cluer.nkn.uidaho.edu

2. Baumgaertner, B. and Research Computing and Data Services. "The Evidence Game: A digital tool for assessing standards of evidence." Supported by Institute for Interdisciplinary Data Sciences, U. of Idaho. Example: https://evidencegame.nkn.uidaho.edu/game/12

3. Baumgaertner, B. (second draft in progress) *Decision Theory: A Conceptual Introduction* https://bertybaums.github.io/decisiontheory/

4. Baumgaertner, B. and O'Rourke, M. "Critical Thinking Worksite" (available online: https://www.webpages.uidaho.edu/bbaum/crit_think/index.htm)

5. **Baumgaertner, B.**, *IntroLogic*, a YouTube channel with supplementary video lectures, total 68k views as of Oct 15, 2023: https://www.youtube.com/user/IntroLogic/

UNPUBLISHED

1. Baumgaertner, Bert (under review) "Only the Story Completes the Grid: Hand-Authored and Grammar-Generated Puzzles for Narrative Augmented Reasoning" *ICLR 2027*. Sole author.

2. Antia, R., Baumgaertner, B., Bull, J.J. (2025) "Density dependence affects population recovery following an epidemic" bioRxiv preprint. https://doi.org/10.64898/2025.12.12.693954

3. Jason Starace, Bert Baumgaertner, Terence Soule. "Deceptive Algorithm Research Must Be Encouraged and Governed by What It Enables, Not by Who Conducts It" (submitted to NeurIPS)

4. Jason Starace, Bert Baumgaertner, Terence Soule. "Ethical Implications of Training Deceptive AI" arXiv:2604.03250

5. Baumgaertner, Bert (submitted) "Metaphor in the Machine: From Source to Target Domains without Dogmas about Conceptual Structure"

6. Baumgaertner, Bert; Zhan, Boyu; Sheneman, Luke; Brunsfeld, John; Culbertson, Jason*; Barzee, Skyler*; Slack, Clayton* (in preparation) "Measures of Intelligence We Live By: Metaphors, Representation-Sensitivity, and 'Program' Synthesis"

7. Baumgaertner, B. and Molyneux, B. (in revision) "Complexity and Philosophical Cognition"

FUNDING

Crossing the Innovation Valley of Death: Democratizing Data and Artificial Intelligence for Research Administration (AI4RA)
Source: NSF GRANTED program, Award #2427549 (funded; PI at U. Idaho)
Effort: Senior personnel, 2024 – present; course buyout, 2025–26 academic year
Description: AI-ethics and AI-literacy training modules for a multi-institution research-administration community of practice

Developing a Web-based AI Code Names Game to Promote AI Literacy
Source: 2024-2025 CLASS Undergraduate Collaborative Key Fund Award
Budget: $2,962 ($1,481 came from Department match)
Description: Funds support two undergraduates: Ed Hiestand and Owen Carswell

Modeling Turn-Taking in Human-Chatbot Dialogues using Philosophy of Language
Source: 2024-2025 CLASS Summer Research Grant
Budget: $3,660 ($1,200 came from Department match)
Description: Funds support two undergraduates: Grace Clifton and Skyler Barzee

2024 National Institutes of Health Pilot Grant
Role: Collaborator (Senior Personnel)
PI: Florian Justwan
Source: NIH, facilitated by Institute for Modeling Collaboration and Innovation
Budget: $115,567
Description: Disease Dynamics and Standards of Evidence

2023 Data Access Grant
Source: Institute for Modeling Collaboration and Innovation
Budget: $3882
Description: Standards of Evidence. A pilot survey for preliminary results for an NSF proposal.

2022-2023 Modeling Access Grant
Source: Institute for Modeling Collaboration and Innovation
Budget: Access to IMCI post-doc at 20% effort for 4 months (about 3.2 weeks effort, October 10, 2022 - February 9, 2023).
Description: Standards of Risk in a Behavior-Disease Compartmental Model

2022-2023 Undergraduate Collaborative Research Key Fund Award
Source: College of Letters, Arts, and Social Sciences
Budget: $ 1,477
Description: collaborrative research grant with undergraduate on the project "Standards of Evidence and Political Misperception"

2022-2023 University of Idaho Think Open Fellow Award
Source: University of Idaho Library
Budget: $1,500
Description: Deploy open access Decision Theory text

2021-2022 University of Idaho Think Open Fellow Award
Source: University of Idaho Library
Budget: $1,500
Description: Develop open access Decision Theory text

Future directions for the study of echo chambers: a student-driven white paper
Source: 2021-22 CLASS Undergraduate Collaborative Research Key Grant (U.Idaho)
Budget: $1,500 (2021-22)
Description: Supervise Trevor Woodward's writing of a white-paper on misinformation and the backfire effect

Disagreement: From Theory to Practice and Back
Role: Principal Investigator
Source: University of Idaho Seed Grant
Budget: $11,427 (2018-2019)
Description: This proposal uses semi-structured workshops to assess how well theories of disagreement (as established in the philosophical literature) apply to concrete issues.

Theory, practice, and social aspects of reproducible science
Role: Co-Principal Investigator
Source: University of Idaho Office of Research and Economic Development
Budget: $ 40,000 (2016-2017)
Description: Many factors affect the nonreproducibility of experimental findings, but relative contribution of these factors and complex interactions among them remain unquantified and largely unknown. Efforts to address the reproducibility crisis have a better chance of success for improving scientific practice if first a theory of nonreproducibility is advanced.

Simulating signals and security
Role: Principal Investigator
Source: NSF award DBI0939454 for BEACON Center for the study of Evolution in Action
Budget: $97,707 (2015-2016)
Description: The project is to evolve signaling systems *in silico* in order to understand the conditions under which compositional signals can arise. We implement and test the theory by developing a signaling system for Active Authentication.

Agent-based models of viral co-infection
Role: Project Director
Source: National Institutes of Health award P20 GM104420
Budget: $126,552/yr (2015-2018).
Description: Model social and behavioral aspects of epidemiology, with a focus on co-infections. This is one of three projects in the new Center for Modeling Complex Interactions at University of Idaho. Involves UI CLASS commitment for new tenure-track line in Department of Philosophy.

Opinion dynamics and the race against climate change
Role: Principal Investigator
Source: Kurt Olsson Early Career Research Fellowship, College of Letters, Arts, and Social Sciences at University of Idaho
Budget: $11,241 (2014-2015)
Description: Model the spread of opinions with a focus on local interactions. The aim is to better understand the processes that govern how opinions shift in a population, particularly when individuals have a small confirmation bias that amplifies their beliefs.

Awards

2026 NEH Summer Institute for Higher Education Faculty — Self-Knowledge for Humans and Artificial Systems
- University of California, Berkeley, June 14–26, 2026. Competitive national selection with participant stipend. Directors: Alva Noë, Christian Coseru, Evan Thompson.

2026 Student Success Symposium Ignite Award — Innovation in Teaching for Persistence
- Center for Excellence in Teaching and Learning, University of Idaho

University of Idaho Presidential Mid-Career Excellence Award
- Awarded Spring 2023, with a prize of $10,000
- "As a recipient of this award, you are considered as one of the university's most gifted faculty members who continues to serve as a role model, a source of inspiration for students, and whose scholarship contributes to the intellectual development and lives of people in Idaho and globally."

Alumni Award for Excellence
- University of Idaho award for excellence in advising, 2021

Outstanding Graduate Student Teaching Award
- Graduate studies award for excellence in teaching at UC Davis ($250), 2013

The Michael V. Wedin Teaching Award
- Teaching award for excellence in graduate student teaching at UC Davis, 2012

Social Sciences Dean's Doctoral Fellowship for Excellence
- Research award for doctoral students at University of California, Davis, whose record of achievement demonstrates exceptional promise in research and scholarship ($1,000), 2012

Selected Presentations

**Peer-reviewed *Invited

1. ** (Poster Presentation) "Measures of Intelligence We Live By: Metaphor, Representation, and Public AI Literacy", Philosophy of Science Association 30th Biennial Meeting, San Diego, CA, November 19–22, 2026.

2. "Your Risk-Averse Mindset Needed: Responsibly Navigating the AI Adoption Rodeo" (with Nathan Wiggins), NCURA Region VI/VII Regional Meeting (RM26), Denver, CO, November 11, 2026.

3. "Thinking Well with AI" in the workshop "Augmenting Research Administration with AI" (with Nathan Layman), NCURA Region VI/VII Regional Meeting (RM26), Denver, CO, November 8, 2026.

4. ** (non-presenting co-author) "Argue with a Chatbot? Using Artificial Intelligence to Support Scientific Thinking and Reasoning" (with Yin Hong Cheah and Zeth duBois) 125th convention of the School Science and Mathematics Association, Chicago IL November 5-7, 2026

5. * "Generative AI in Research and Academia: Ethical Considerations and Practical Applications", Postdoc Appreciation Week, University of Idaho, September 24, 2026.

6. * "How AI Spotlights Science's Narrative Blind Spot", Mini Symposium on AI in Science, School of Philosophy, Psychology and Language Sciences, University of Edinburgh, August 26, 2026.

7. ** "Post-Hoc Understanding of Metaphor Processing in Decoder-Only Language Models via Conditional Scale Entropy" (with L. Chakrabarti, J. Johnson-Leung, A. Vakanski, M. Xian, B. Zhang), 4th Workshop on Explainable and Ethical AI, International Conference on Pattern Recognition (ICPR 2026), Lyon, France, August 17–20, 2026.

8. "Is a Self a Prerequisite for Solving Puzzles in the Narrative Augmented Reasoning Corpus?", participant presentation, NEH Summer Institute on Self-Knowledge for Humans and Artificial Systems, University of California, Berkeley, June 25, 2026.

9. * "Was the Gettier Problem Inevitable? Language Models as Instruments for Complementary Inquiry", Department of Philosophy, Boise State University, April 25, 2026.

10. * "New Trails, Old Coastline: Humanities Takeover in the Era of Agentic AI", public humanities lecture, Boise State University, April 24, 2026.

11. * (Panelist) AI and the arts, Lionel Hampton School of Music, University of Idaho, April 14, 2026.

12. * "Metaphor in the Machine: From Source to Target Domains without Dogmas about Conceptual Structure" Pacific American Philosophy Association, April 12, 2026

13. * "Creating a Network of Interdisciplinary Human-AI Collaborators" Innovate Idaho 2026 — AI and Open Education Symposium, April 10, 2026. Link

14. * "The Dark Room and the Ruthless Curiosity it Demands: Graduate Research in the Age of Agentic AI", invited speaker to ENVS 5010 Seminar series, College of Natural Resources, University of Idaho, March 23, 2026

15. "A Formal Framework for How Inferential Commitments Create Deep Disagreements" Computational Social Philosophy Seminars, March 23, 2026

16. **"Metaphors We Intend With: Teaching AI Alignment Through Puzzles" AI and Data Ethics: Philosophy and Pedagogy, University of Southern California, March 6-7, 2026

17. **"Bots as foils: Real students learn by teaching AI students" University of Idaho Student Success Symposium: Ignite and Elevate, February 27, 2026

18. "How will our ability to access PhD++ expertise at $20 a month impact 'modeling collaboration and innovation'?" Institute for Modeling Collaboration and Innovation Brown Bag Lunch Series, February 23, 2026

19. *"The Grasshopper's Utopia: AI, Games, and the Future You're Playing Into" Vandal Gaming Convention, University of Idaho, Feb 28, 2026

20. *"MARC Your Human Intelligence" University of Idaho POP Talks, October, 2025. Press Release and Video

21. * "EASY for Humans, HARD for AI: Metaphor Abstraction and Reasoning Corpus" Malcolm Renfrew Interdisciplinary Colloquium, U. of Idaho, September 2, 2025.

22. * "Chatbots and Misinformation: The Semantic Turn, for Better and Worse" Institute for Modeling Collaboration and Innovation - Brown Bag Lunch Series, U. of Idaho, January 13, 2025.

23. * "AI Literacy - What is and isn't working" Institute for Interdisciplinary Data Sciences - AI Literacy Series, U. of Idaho, November 7, 2024.

24. *"AI Literacy for Everyone Who Learns and Inquires" The Renfrew Colloquium, joint presentation with Casey Johnson, U. of Idaho, October 22, 2024.

25. (non-presenting co-author) "Standards of evidence: Disease Dynamics and Human Evidence Gathering" University of Idaho, October 1, 2024

26. * "LLMs, Bias and Ethical Dilemmas" joint lecture with Casey Johnson in Boyu Zhang's graduate course on Natural Language Processing, April 29, 2024.

27. * "AI Literacy – A Discussion Spark in Using AI in Critical Thinking" Generative AI in Higher Education in Idaho, April 19, 2024.

28. * "Humanities Takeover: Values and the work towards AI literacy" Institute for Interdisciplinary Data Sciences Seminar Series, April 18, 2024.

29. * "Agent-Based Models and Generative Explanations in Complex Systems" University of Idaho Library Tech Talks, March 28, 2024

30. * (Panelist) "AI Generative Text Tools: Changing Academia since 2023", Malcolm Renfrew Interdisciplinary Colloquium, University of Idaho, August 22 & 29, 2023.

31. ** "Risk Tolerance, Norms, and Epidemic Waves", 6th International Conferene in Applied Mathematics, Modeling and Computational Science, Wilfrid Laurier University, Waterloo, Ontario, Canada, August 14-18, 2023.

32. ** "Risk Tolerance, Norms, and Epidemic Waves", Behavioral Epidemiology Minisymposium, CAIMS, (virtual) June 14, 2023.

33. ** "Precedent and Interpersonal Convergence in the Method of Reflective Equilibrium", Philosophy of Science Association Biannual Meeting, Pittsburgh, PA, November 10-13, 2022.

34. ** "Convergence and Shared Reflective Equilibrium", Reflective Equilibrium 51 Years after A Theory of Justice, University of Bern, Switzerland, May 12-14, 2022.

35. ** "Openness and Reproducibility from a Model-Centric View", Philosophy of Science Association Biannual Meeting, Baltimore, November 11-14, 2021

36. * "Perspectives on Disagreement – When to dig in and when to compromise?" (2 hour workshop) Effective Communication Skills for Applied Mathematicians, https://ecsam.ca/module-5/, June 3, 2021

37. * "Representation and Modeling of the COVID Outbreak", panelist in Symposium on Ethics of COVID-19 Science, March 19, 2021

38. * "Motivated Disagreement and Vaccination Decisions" Project ECHO, a virtual forum connecting Idaho's rural health care workforce for continuing education and collaboration, March 16, 2021

39. *"Scientific Discovery, Reproducibility, and Open Science" with Berna Devezer, Renfrew Interdisciplinary Colloquium, University of Idaho, February 5, 2019

40. ** "Model and Theory Development from an Epidemiological Perspective", Philosophy of Science Association Biannual Meeting, Seattle, WA, November 1-4, 2018.

41. ** "The Backfire Effect in Opinion Dynamics" Computational Modeling in Philosophy, LMU, Munich, Germany, June 22-23, 2018.

42. "How Values, Trust, and Socio-Political Context Shape Vaccination Attitudes" American Association for the Advancement of Science Pacific Division, Pomona, CA, June 12-15, 2018.

43. * "Disagreement", Keynote Address at The Confluence Project (Youth Water Summit), North Idaho College, Couer d'Alene, Idaho, May 22, 2018.

44. * "Perspectives on Disagreement", Keynote Address at OUR Gem Symposium, Couer d'Alene, Idaho, November 14, 2017.

45. * "The Philosophy and Science of Modeling Infectious Disease", The Annual Norbert Wiener Address on Science and Ethics, South Connecticut State University, March 30, 2017.

46. ** "Spatial Opinion Dynamics and the Effects of Two Types of Mixing", International Congress on Agent Computing, George Mason University, Fairfax, Virginia, November 29-30, 2016.

47. (Poster Presentation) "Some Philosophical Challenges Concerning the Inclusion of Social Norms in Epidemiological Models", Philosophy of Science Association Biennial Meeting, Atlanta, Georgia, November 3-5, 2016.

48. * "How too much structure inhibits the marketplace of ideas", Breaking Barriers: Idaho Local Government Economic Development Summit, Boise, Idaho, October 11, 2016.

49. * "Disjunctive Signaling Games", University of Edinburgh, Scotland, July 8, 2016.

50. * "Mathematical and Computational Methods in Philosophy", Computational Ecology Research Group, University of British Columbia, Okanagan, May 26, 2016.

51. ** "Spatial dynamics of mathematical models of opinion formation", Computationally Assisted Discovery and Experimental Mathematics, Western University, London, Canada, May 12-15, 2016.

52. * "Opinion strength influences the spatial dynamics of opinion formation", Spring Colloquium Series, Department of Philosophy, Simon Fraser University, January 29, 2016.

53. ** "Imperfect Observations Produce Asymmetric Signaling Roles", Northwest Philosophy Conference, North Idaho College, Coeur d'Alene, Idaho, October 9-10, 2015.

54. ** "Belief Amplification and Imitation in an Extended Voter Model", Agent-Based Modeling in Philosophy, LMU, Munich, Germany, December 11-13, 2014.

55. * "Modeling Opinion Dynamics about Climate Change", Malcolm M. Renfrew Interdisciplinary Colloquium, University of Idaho, November 4, 2014.

56. ** "The Importance of the Environment in Complex Modeling Practices: a Response to Craver and Bechtel" (with Sarah Roe), Western Canadian Philosophical Association 2014 Annual Meeting, Vancouver, British Columbia, October 3-5, 2014.

57. ** "A Limiting Case for the New Mechanists: Intra-level Causation is Insufficient for Modeling Complex Biological Systems" (with Sarah Roe), Philosophy of Biology at Madison, Madison, Wisconsin, May 30-June 1, 2014.

58. ** "Agent-Based Models as Mixed-level: Lessons from *E.coli*", Conference on Reduction and Emergence in the Sciences, Munich, Germany, November 14-16, 2013.

59. * (Poster Presentation) "Agent-Based Models as Mixed-level: Lessons from *E.coli*", Annual IBEST Science Expo, University of Idaho, Moscow, Idaho, October 18, 2013.

60. ** "Reflective Inequilibrium", 65th annual Northwest Philosophy Conference, Pacific University, Forest Grove, Oregon, October 4-5, 2013.

61. * "Agent-Based Models as Mixed-level: Lessons from *E.coli*", IBEST Science Update, University of Idaho, Moscow, Idaho, September 12, 2013.

62. * "Millian Mingling: Simulating Millian Ideas on Diversity and Measuring its Epistemic Benefits", Invited Presentation, University of Alaska, Anchorage, Alaska, 2013.

63. * "Millian Mingling: Simulating Millian Ideas on Diversity and Measuring its Epistemic Benefits", Invited Presentation, University of Idaho, Moscow, Idaho, 2013.

64. * "Millian Mingling: Simulating Millian Ideas on Diversity and Measuring its Epistemic Benefits", Invited Presentation, Flagler College, St. Augustine, Florida, 2013.

65. ** "An Agent-Based Model of Millian Mingling", Copenhagen-Lund Workshop in Social Epistemology, Lund, Sweden, 2012.

66. "Millian Mingling: How Valuable is Epistemic Diversity?" Graduate Student Workshops, University of California Davis, 2012.

67. * "Reflective Inequilibrium", Wilfrid Laurier University, 2012.

68. * "Cognitive Science and Vagueness", First CSLI Workshop on Logic, Rationality, and Interaction, Stanford University, 2012.

69. * "A Defense of Modeling Vagueness with Discrete Degrees", Fourth Workshop on Philosophy of Information, University of Hertfordshire, 2012

70. ** "Predicates and their Interpretations in Natural Reasoning", Conference on Computing Natural Reasoning, Tilburg, the Netherlands, 2011.

71. ** "Modeling Echo Chambers", Copenhagen-Lund Workshops in Social Epistemology, Lund, Sweden, 2011.

72. * "Characterizing Vagueness: Dynamics and Ignorance", Vagueness Circle, Institute for Logic, Language, and Computation, University of Amsterdam, 2011.

73. ** "The Problem of Vagueness from a Philosophy of Information Perspective", Philosophy of Engineering and Artifact in the Digital Era (PHEADE), Bukovina, Romania, 2011.

74. ** "Reflective Inequilibrium", International Association for Computing and Philosophy (IACAP), Aarhus, Denmark, 2011.

75. **"Adaptive Concepts: Using Frames to Solve a Problem of Vagueness", Canadian Philosophical Association, Montreal, Canada, 2010.

76. Commentary on David Barnett's "Vague Entailment", American Philosophical Association, Pacific Division, San Francisco, USA, 2010.

77. ** "Using Frames to Solve the Problem of Vagueness", Second Conference on Concept Types and Frames, Heinrich Heine Universität, Düsseldorf, Düsseldorf, Germany, August 24, 2009.

78. ** "Vagueness in Computational Linguistics: Contextualist Models are Insufficient", European Conference on Computing and Philosophy (ECAP), University for Science and Technology, Montpellier, France, June 16-18, 2008.

79. "Introduction to Philosophy of Language", Guest Speaker (invited by D. Koegler), Waterloo Collegiate Institute, May 20, 2008.

80. ** "Vagueness", Western Research Forum, the University of Western Ontario, March 29, 2008.

81. "Rules in Linguistic Research: From Skepticism to Modest Realism", PGSA Colloquium Series, Department of Philosophy, the University of Western Ontario, February 6, 2008.

ACADEMIC EXPERIENCE

Research Supervision

* indicates member of underrepresented group

AI Ethics and Inquiry Lab (co-lead with Casey Johnson, Ass. Prof. of Philosophy)

Jason Culbertson, 2024-present (Philosophy, Computer Sci., U. Idaho)
Skyler Barzee*, 2024-present (Philosophy, U. Idaho)
Grace Clifton, 2024-2025 (Political Science, U. Idaho)
Karina Sande*, 2024-2025 (Philosophy, U. Idaho)
Ed Hiestand*, 2024-present (Philosophy, U. Idaho)
Owen Carswell, 2024-present (Philosophy, U. Idaho)

BAFL (Bert and Florian Lab - Florian Justwan, Ass. Prof. of Political Science))

Nelson Abel, 2025-present (Politics and Philosophy, U. Idaho)
Maxwell Leisinger, 2025-present (Philosophy, Mathematics, U. Idaho)
Jacob Bindley, 2024-25 (Politics and Philosophy, U. Idaho)
Alexa Benitez*, 2024-25 (Politics and Philosophy, U. Idaho)
Karina Sande, 2024-25 (Politics and Philosophy, U. Idaho)
Kira Haile, 2024-25 (Politics and Philosophy, U. Idaho)
Bergen Kludt-Painter, 2024-25 (Politics and Philosophy, U. Idaho)
Dylan James, 2023-24 (Politics and Philosophy, U. Idaho)
Kendal Mitton, 2022-24 (Politics and Philosophy, U. Idaho)
Emilee Adams*, 2022-24 (Politics and Philosophy, U. Idaho)

Postdoctoral

Chenangnon Tovisode*, 2022 - present (IMCI, U. Idaho)
Aniruddha Belsare*, completed 2018 (CMCI, U. Idaho)
Luis Gustavo Nardin, completed 2017 (CMCI, U. Idaho)

Graduate

(Major Professor) Victoria DePalma, PhD 2019 (Environmental Sciences, U. Idaho)
Kelly Christensen*, (Math, U. Idaho)
Kristen Petersen*, graduated on time, 2016 (Statistics, U. Idaho)
Joseph DeAguero*, (Bioinformatics & Computational Biology, U. Idaho)
Jeannie Matheison (Environmental Sciences, U. Idaho)
Jimit Majmudar*, graduated on time, 2016 (Math at UBC Okanagan)

Undergraduates (not associated with labs)

Trevor Woodward, 2019 - 2022 (Philosophy, U. Idaho)
Garret Caudle, graduated 2019 (Philosophy, U.Idaho)
Nova Tebbe, graduated 2019 (Philosophy and Biology, U. Idaho)
Camden Clark, graduated 2019 (Computer Science, U. Idaho)
Emma Bateman, graduated 2018 (Computer Science, U. Idaho)
Emma Carson, graduated 2018 (Political Science, U. Idaho)

Jordan Kizer, graduated 2018 (Poltical Science, U. Idaho)
Dorothy Catey*, graduated on time, 2018 (Math, U. Idaho)
Peter Fetros, graduated on time, 2018 (Engineering, U. Idaho)
Camden Clark, graduated on time, 2018 (Computer Science, U. Idaho)
Cody Briston, graduated on time, 2016 (Philosophy, U. Idaho)
Cassie Greenwald*, graduated on time, 2016 (Philosophy, U. Idaho)
Crystal Hernandez*, graduated on time, 2016 (Philosophy, U. Idaho)
Kaitlyn Hanks, graduated on time, 2016 (Environmental Sciences, U. Idaho)
Jennifer Stephenson, graduated on time, 2016 (Environmental Sciences, U. Idaho)
Zach Hull, graduated on time, 2015 (Environmental Sciences, U. Idaho)
Brendan McConnor, graduated on time, 2015 (Environmental Sciences, U. Idaho)

Committee Reading Member
* indicates member of underrepresented group

Doctorate

Torrey Mortenson, in progress (Biological Engineering, U. Idaho)
Malcom Rupert, 2017 (Math, U. Idaho)
Siavash Riazi*, 2016 (Bioinformatics & Computational Biology, U. Idaho)
Jesse Oldroyd, 2016 (Math, U. Idaho)
Marcie Galbreath, 2015 (Education, U. Idaho)

Masters

Avneet Kaur, Masters, in progress (Mathematics, UBC Okanagan)
Steven Alberding, Masters 2023 (Mathematics, U. Idaho)
Noah M. Marshall, 2020 (Mathematics, UBC Okanagan)
Sarah Hamilton, 2019 (Mathematics, UBC Okanagan)
Danielle Gentry, 2016 (Environmental Sciences, U. Idaho)
Brett Miller, 2015 (Environmental Sciences, U. Idaho)

Courses Taught (regular curriculum)
† indicates upper division/graduate level course

† *Philosophy of Science*

- University of Idaho, Spring 2014, Fall 2016, Spring 2022
- University of California, Davis, Summer Session II 2010

† *Metaphysics*

- University of Idaho, Fall 2021, 2023

† *Theory of Knowledge*

- University of Idaho, Spring 2021

† *Intro to Decision Theory*

- University of Idaho, Fall 2020, 2022

† *John Stuart Mill (Senior Seminar)*

- University of Idaho, Spring 2020

† *History of Modern Philosophy*

- University of Idaho, Spring 2019, 2020

† *Thought Experiments, Simulations, and Models (Special Topics)*

- University of Idaho, Fall 2018

Introduction to Symbolic Logic

- University of Idaho, annually since 2013
- University of California, Davis, Winter 2012

† *Philosophy of Mind*

- University of Idaho, Fall 2018

† *Philosophical Methods (Senior Seminar)*

- University of Idaho, Spring 2018

Philosophy of Alcohol

- University of Idaho, Spring 2017

† *Philosophy of Evolution (Senior Seminar)*

- University of Idaho, Spring 2015

† *Philosophy of Language*

- University of Idaho, Fall 2014, Spring 2019

† *Environmental Philosophy*

- University of Idaho, Fall 2013, 2014, 2015

Belief and Reality

- University of Idaho, Spring 2014, 2015

Biomedical Ethics

- University of British Columbia (Okanagan), Summer 2014

Critical Reasoning

- University of Idaho, Fall 2021, Spring 2022
- University of California, Davis, Summer Session II 2009

Directed Studies
All directed studies are upper division

BAFL Research Methods (Bert and Florian Lab)

- University of Idaho, Fall 2023

Non-Classical Logic (including Modal Logic)

- University of Idaho, Fall 2016, 2017, 2020, 2023, Spring 2017, 2018, 2019, 2022

Philosophy of Math

- University of Idaho, Fall 2020

Relevant Logic

- University of Idaho, Spring 2021

Metalogic

- University of Idaho, Spring 2014

Opinion Dynamics

- University of Idaho, Spring 2015

Philosophy of Perception

- University of Idaho, Fall 2015

Selected Service

Academic Committees

- AI Strategic Plan Execution Group, University of Idaho, 2025 – present
- AI Literacy Task Force, University of Idaho, 2025 – present
- Outside Member, Third-Year Review Committee, Department of Curriculum and Instruction, Spring 2025
- University of Idaho AI Steering Committee, Spring 2024 - 2025
- Artificial Intelligence & Machine Learning Task Force, 2023 - 2024
- "AI at UI" Faculty Study Group, 2023 - 2025
- Chair, Department Promotion and Tenure Committee (Dr. Quinn), Fall 2022
- Member, Department Promotion and Tenure Committee (Dr. Johnson), Fall 2022
- Member, Department Promotion and Tenure Committee (Dr. Hubbs), Fall 2022
- Member, CLASS College Promotion and Tenure Committee, Fall 2021
- Member, COVID Advisory Committee, U. Idaho, Spring 2021
- Voting Member, University Curriculum Committee, University of Idaho, Fall 2018-2021
- Chair, Program Review Committee, North Idaho College, Spring 2017
- Member, Dept. Chair Search Committee, Department of Politics and Philosophy, U. Idaho, Spring 2016
- Member, Hiring Committee (Science & Ethics), Philosophy, U. Idaho, 2016-17
- Member, Hiring Committee (Social/Political), Philosophy, U. Idaho, 2016-17
- Member, Hiring Committee (Political Science), Department of Politics and Philosophy, U. Idaho;
- Member, Tenure Committee for Alex Woo, Department of Mathematics, U. Idaho, Spring 2016
- Head, Instructor Hiring Committee, Department of Philosophy, U. Idaho, Fall 2015
- Member, Curriculum Committee, CLASS, U. Idaho, 2014-2015
- Member, Promotion and Tenure Committee, CLASS, U. Idaho, 2013-2014

Conference/Workshop Organization and Outreach Service

- Committee member, Values & Social Aspects of Science, Philosophy of Science Association, 2025-26
- Idaho AI Catalyst, representing U. Idaho (Idaho State Board of Education appointed), Fall 2025-present
- Co-presenter, Idaho AI Catalysts statewide webinars: AI research tools (January 2026); AI, water, and energy (March 2026)
- Co-lead instructor, Faculty AI Literacy Bootcamp, University of Idaho, June 2026
- Designed the ethics deliberation instrument for *Building Tribal AI Sovereignty*, a University of Idaho workshop for tribal nations of the Pacific Northwest, July 8–9, 2026
- AI reasoning-puzzle outreach session, Moscow Middle School, May 2026

- Careers in the Humanities, co-Organizer and MC, Oct 17, 2023
- Consulting for Extension Collaborative on Immunization Teaching & Engagement (EXCITE), 2021-2022, Contact: Lindsey McConnell-Soong
- Board Member, Efficient Public Collaborative, Idaho, May 2019 - 2021
- Conference co-organizer, Reasoning Well in the Anthropocene: Philosophical Insights on Climate Change (Inland Northwest Philosophy Conference), University of Idaho and Washington State University, April 14-16, 2016
- Conference co-organizer, Philosophy of Science: Inside and Out (18th Inland Northwest Philosophy Conference), University of Idaho and Washington State University, July 31 - August 1, 2014

Conference Commentator / Chair

- Chair at Philosophy of Science Association Conference, November 2018
- Chair at Berkeley-Stanford-Davis Philosophy Conference, April 2012
- Chair at Pacific American Philosophical Association Conference, April 2012
- Commentary on David Barnett's "Vague Entailment", American Philosophical Association, Pacific Division, San Francisco, USA, 2010.

Related Academic Activities

Research Invitations

University of Amsterdam **Fall 2011**

- Research fellowship at the Institute for Logic, Language, and Computation under the supervision of Prof. Dr. Frank Veltman. Studied computational approaches to philosophical questions in mind (vagueness), semantics and pragmatics, and social phenomena (e.g. judgment aggregation). Collaborated with Raquel Fernández (see Baumgaertner, B., Fernández, R., Stone, M. (2012))

Rutgers, The State University of New Jersey **Summer 2011**

- Invited internship to work on "Collaborative Reference in Open Domains" (Prof. M. Stone). Developed a parsing algorithm to handle a grammar with "relaxed" lexical meanings (see Baumgaertner, B., Fernández, R., Stone, M. (2012)).

Training and Workshops

- Equity Leadership Institute, CLASS, University of Idaho, August 2022
- Learning and Gaining from your Model: A Course in Analysis, Application, and Publication of Individual/Agent-Based Models, Technische Universität Dresden, June-July 2016
- NIMBioS Tutorial - Game Theoretical Modeling of Evolution in Structured Populations, University of Tennessee, Knoxville, April 2016
- Write Winning Grant Proposals (by the Grant Writer's Seminars and Workshops in association with the Idaho INBRE Program), Coeur D'Alene, October 2015
- Business for Scientists - Leading and Sustaining your Research Program, Institute for Bioinformatics and Evolutionary Studies, University of Idaho, June 2015
- Restorative Practices and Circles Training (two certificates), University of Idaho, April 2015
- Safe Zone Training, Diversity and Human Rights, University of Idaho, Spring 2014
- Agent-Based Modeling in Philosophy, a Ghent-Tilburg Workshop in Spa, Belgium, May 2012.

Research Groups

- Formal Epistemology Reading Group (virtual), Fall 2021 - Spring 2023
  University of California, Irvine
- Center for Modeling Complex Interactions, Spring 2015-2020
  University of Idaho
- ABM (Agent-Based Modeling) Group, Fall 2010 - Spring 2013
  Department of Psychology (Jeff Schank), University of California, Davis
- Philosophy of Biology Lab, Winter 2010 - Spring 2013
  Department of Philosophy, University of California, Davis
- Vagueness Circle, Fall 2011
  Institute for Logic, Language, and Computation, University of Amsterdam

TECHNICAL SKILLS

Programming in R, NetLogo (and now latent: some LISP, Java and Python)
My research includes the programming of agents and societies.
Typesetting in LaTeX
I have experience in creating sophisticated documents and presentations in LaTeX.

AFFILIATIONS

Philosophy of Science (PSA)
American Philosophical Association (APA)
Erdös #: 4 (Paul Erdös to Robert J. McEliece to Dilip V. Sarwate to Stephen M. Krone)

LANGUAGES

English: Fluent, Spoken and Written
German: Fluent, Spoken and Written (though less practice in the latter)

References

Cailin O'Connor
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Todd Jones
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University of Nevada, Las Vegas
Phone: 702-895-4691
Email: tjones@unlv.nevada.edu

Charles Lassiter
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Holly Wichman
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University of Idaho
Phone: 208-885-7805
Email: hwichman@uidaho.edu

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